

Blueprint: Bernoulli regular route
(flt-regular-extended)

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Abstract

`flt-regular-extended` certifies FLT for the *regular* primes $17 \leq q < 350$ by a cheap route: a kernel Bernoulli-numerator check discharges regularity, and `flt-regular` supplies FLT for regular primes. It reuses the shared Bernoulli/Herbrand base of `flt-cyclotomic-nt` (external leaves).

0.1 The Bernoulli regular route

Definition 0.1.1 (FLT for regular primes (external)). *flt-regular's theorem: a regular prime satisfies FLT (via Kummer's ideal-descent argument).*

Definition 0.1.2 (Bernoulli / Herbrand base (external)). *The kernel Bernoulli check `not_irregular_of_bModN` and Herbrand's theorem, from `flt-cyclotomic-nt` — the algebraic direction “Bernoulli-regularity $\Rightarrow p \nmid h$ ”.*

Theorem 0.1.3 (Regularity check $\Rightarrow$ FLT). *A prime passing the regularity check satisfies `FermatLastTheoremFor`, the generic bridge instantiated per prime.*

Theorem 0.1.4 (Bernoulli-regularity $\Rightarrow$ FLT). *If p divides the numerator of none of $B_2, \dots, B_{p-3}$ (a kernel-**decided** test), then p is regular and FLT holds for p .*

Theorem 0.1.5 (First regular prime past 100). *`FermatLastTheoremFor` 107, the first regular exponent above 100. All 47 regular primes in $[17, 350)$ are proved by this same route (files `Flt17.lean` ... `Flt349.lean`).*